

# **BAN SYNDROME AND ITS DESTABILISATION EFFECT : EVIDENCE FROM INDIAN AGRICULTURE**

## **COMMODITY MARKET**

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### **ABSTRACT**

The abrupt suspension of agriculture commodities from futures trading has become a characteristic feature of many emerging economies especially India and China. The new debate is whether such an extreme step of a ban, a viable solution to the overall development of commodity markets or is it a faux pas to address the apprehensions of policy makers who believe that agriculture commodity prices are being rigged by speculators. The aim of the study is to explore the impact of such abrupt suspension on spot return volatility and the price discovery mechanism during pre-ban and post relaunch phases of all 14 cases of commodities ban (tur, urad, sugar, wheat, rice, guarseed, guar gum, castor seed, rubber, chana, potato, soya refined oil) from 2003 to 2017. The study also examined destabilization impact of futures trading by examining the impact of unexpected trading activity of futures market (volume traded, open interest and value traded) using Hodrick Prescott filter. The study found average spot return volatility has risen during ban phase as compared pre ban and post ban phase and destabilization impact was found to be significant for post ban futures trading implying that the decision to ban these commodities was an attempt in wrong direction.

### **KEYWORD**

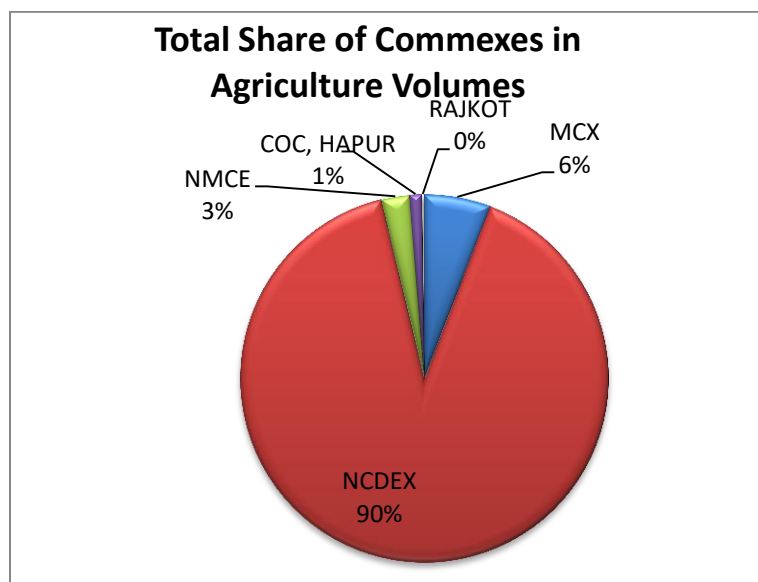
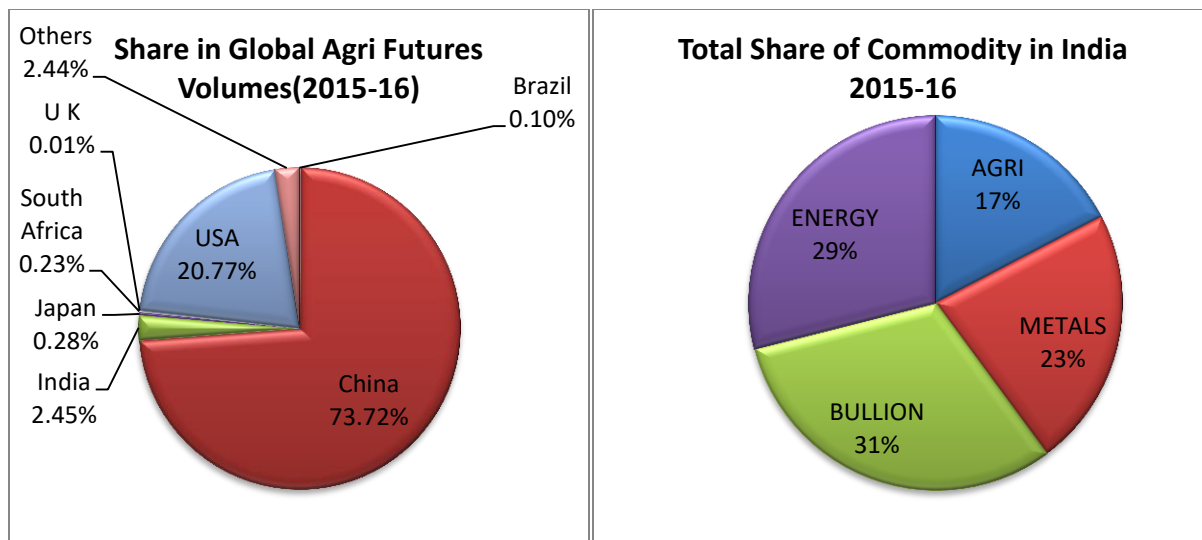
**Agriculture commodity, Ban syndrome, Destabilisation Effect, Price Discovery, India**

## INTRODUCTION

There is a renewed public opinion on the long standing debate of the desirable level of government regulation and heat is turning against the dominant view that that government regulation improves the inefficiency of private sector and protects against negative externalities. Market oriented reforms have started to garner public support and a glaring evidence of it comes from the distortion caused by the ban syndrome to Indian agriculture commodity derivative markets. The Indian agriculture spot markets have been trapped in the quagmire of a gamut of restrictions ranging from price controls like minimum support price to stock limits leading to a disaggregated spot markets. With these dilapidated agriculture spot market conditions, one ray of hope comes from the notion that can futures market pull up the spot markets and correct its distortions. But unfortunately, this also, could not be achieved in India, despite the existence of a long trading history of organized agriculture commodity derivative markets since 1875 owing to continuous bans by government on agriculture commodities.

The abrupt suspension of agriculture commodities from futures trading has become the characteristic feature of many emerging economies like India and China. Commodity derivative trading in India started with cotton exchange in Bombay and oilseeds and jute in started trading. However from mid 1930s because of shortage in essential commodities, futures trading were restricted. Taking the support of various legislations like Cotton Prohibition Act, 1939 and Defence of India Act, 1935 etc. futures trading was banned for cotton, oilseeds and food grains in 1939. After independence, Forward Contracts (Regulation) Act, (FCRA) 1952 and Essential Commodities Act, 1955 completely ripped apart the entire agriculture futures trading by imposing a blanket ban on all agriculture commodities except turmeric and pepper for four decades. Several government lead committee reports urged the regulator to re-introduce agriculture commodity trading (Shroff Committee,1950; Dantwala Committee 1956; Khusro Committee,1980; Kabra Committee,1994) by stressing the benefits of futures trading in agriculture. But major big bang reforms in Indian agriculture commodity derivatives markets came in 2003 when 3 national multi-commodity exchanges were set up and trading was permitted in 113 commodities for the first time. Since then 14 commodities (Table 1) have been suspended on apprehensions by policy makers who believed it to be handiwork of speculators rigging the market.

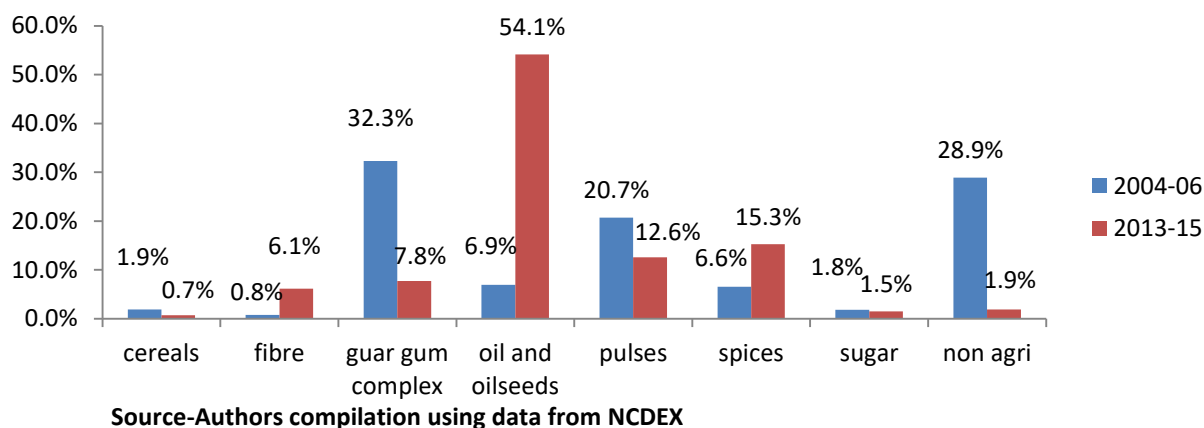
Mapping our positions in global agriculture commodity derivative markets, Figure 1 portrays a dismal picture with India generating only 2.45% of trading volumes in agriculture as compared to China (74%) and U.S.(21%) owing to 4 decades lost due to blanket ban of agri trading.. Analysing the importance of agriculture as a commodity amongst other asset classes, SEBI (2016) reports that only 17% of futures volumes owe to agriculture despite India being the largest producer and exporter of agriculture commodities, and largest volumes comes from Bullion and Energy segment in 2015-16. There are 3 registered national commodity exchanges in India and 3 regional exchanges, Figure 3 shows that National Commodity Derivative Exchange (NCDEX) has a monopoly in agriculture derivative trading with 90% volumes being generated and marginal volumes by other exchanges.



Source – Future Industry Association (as accessed on 30<sup>th</sup> December, 2016) & SEBI Annual Report 2015-16.

Of 21 agriculture commodities being traded in India , the composition from NCDEX shows that in 2004-06, on an average 32% of volumes came from Guar Complex(Guar seed , Guargum) but in 2014-16 the centre of gravity shifted to Oil and Oilseed Complex garnering 54% of total volumes traded. The above mentioned macro level descriptive shows that India has not been able to achieve the kind of exponential growth it saw for stock exchanges and forex market due to extensive government regulation that impeded its growth viz-a-viz its trading partners like China and U.S.

**Figure - Share of agriculture commodity trading on NCDEX (in terms of volume)**



Commodity derivatives have been dubbed as a risk management and price discovery vehicle that enables parties to hedge their commodity exposure and allows efficient transmission of information. But this has long been challenged by academia, regulators and policy makers who believe that uninformed speculators distort the price discovery process (Newbery, 1987) and rig the markets as was accused during Global Food Crisis 2007-08. This long standing debate on the impact of introducing futures trading on underlying spot markets has garnered huge attention in academia and policy circles for varying asset classes: stock, bond, currencies, indices and commodities. The rationale behind introducing derivatives for any asset class has been the low transaction cost and minimal margin requirement relative to over the counter markets that helps in investors to hold large highly leveraged positions (IMF, 1998). An equally convincing reason is that the financial derivative instruments deepen the markets and improve its efficiency by infusing much needed liquidity and reining in cash price volatility (Action, 1971; Darrat & Rahman, 1995; Koutyos & Tucker, 1996; Stoll, 1988; Tse, 1998). However, a parallel strand of literature accused these derivative instruments of destabilizing the underlying spot markets by increasing the spot price volatility (Cagan, 1981; Cox, 1976; Figlewski, 1981; Hart & Kreps, 1986; Stein, 1987; Bessembinder & Seguin, 1993; Choi and Subrahmanyam, 1994).

The three critical issues which need to be examined to understand the role of futures trading in agriculture markets are, firstly, whether introduction of futures trading has destabilized spot markets, secondly, if not, has futures trading rendered its twin objective of efficient price discovery and hedging effectiveness and thirdly, is banning the commodity from futures trading a viable solution to the problem of abrupt price spikes in agriculture prices or has it destabilized the spot markets. This study finds answers to the above mentioned issues in the context of Indian agriculture derivative markets.

Dwelling on the first issue of destabilization effect of derivatives, mixed results were obtained from the extant literature encompassing the effects of both futures and options trading on underlying spot markets. Using varying methodology ranging from event study to augmented GARCH models, (Conrad, 1989; Detemple & Jorion, 1990; Klemkosky, 1989; Shastri, Sultan, &

Tandon, 1996; Stucki & Wasserfallen, 1994; Whaley, 1980) found decrease in underlying spot volatility post futures/options were introduced for equity markets. Similar results were obtained for interest rates, currency and indices (Edward, 1988; Ely, 1991; Bessembinder, 1992; Jabbour, 1994; Schwarz, 1991) but contrary results of increase in spot price volatility were found (Clifton, 1985; Kawaller, 1993; Chatrath, 1993; Crain, 1995). Some found evidence for stabilization role of futures and option markets as it leads to speedy transmission of information leading to better price discovery mechanism and risk management (Powers, 1970; Danthine, 1978; Stoll, 1988; Bessembinder, 1993; Bohl, 2009; Lee, 2013). In India too, studies on introduction of futures trading on agriculture commodity markets found mixed results for varying agriculture commodities (Kamara, 1982; Singh & Kumar, 2005; Sahi, 2006; GOI, 2008) found that spot volatility reduced for hessian crop and did not change wheat, turmeric, sugar, cotton, raw jute and soya oil. On the other hand, (Singh, 2000; Yang et al., 2005; Nath, 2008; Sehgal, 2012; 2013; Ahmad, 2015) observed that futures trading had destabilized spot market for urad, wheat and gram.

Summarising the huge body of literature that examined the destabilization effect of futures trading on spot prices, two major strands were observed, with the first gamut of studies attempted to explore the impact on underlying spot price volatility using generalized autoregressive conditional heteroscedastic type augmented models by using dummy variables to differentiate between pre and post futures trading phase (Working, 1962; Bessembinder and Seguin, 1992; Kamara, 1993; Antoniou *et al.*, 1998, 2005; Gulen and Mayhew, 2000; McKenzie *et al.*, 2001; Yang et al. 2005). Second slew of studies attempted to study the dynamic relationship between future trading activity using proxy like daily volume traded, open interest and value traded with underlying spot price volatility (Lee *et al.*, 1999; Cohen *et al.*, 2002; Barber and Odean, 2008; Kaniel *et al.*, 2008) as they believe that the unexpected component of trading activity represent the noise traders that distort the price discovery mechanism of futures markets. Few studies conducted in Indian context (Adrangi and Chatrath, 1998; Raizada and Sahi, 2006; Bhargava and Malhotra, 2007; Paul, 2010; Sehgal et al., 2012; Ahmad, 2015; Sharma, 2015) provided evidence of destabilizing impact of futures trading.

The second issue of whether futures market render the job of efficient price discovery and hedging efficiency has also been extensively researched for varying asset classes and markets. Most notable one being (Chowdhury, 1991; Crowder, 1993; Kellard et al., 1999; Beck, 2006; Campiche, 2007; McKenzie, 2010) used cointegration, error correction and granger causality techniques to explore long run and short term equilibrium between spot and futures prices for commodity markets. In Indian context, (Thomas and Karande, 2001; Naik and Jain, 2002; Kumar, 2004; Praveen, 2006; Ahuja, 2006; Roy, 2008; and Roy and Dey, 2009; Sehgal, 2012) found that futures lead spot in Indian commodity markets and for most agriculture commodities.

The third and key issue which has not been explored in previous studies is the impact of ban on spot price volatility and price discovery mechanism post relaunch of such commodities. As banning a commodity is a big blow to its market efficiency and price discovery mechanism as investor's lose trust and become more fearful in hedging risk for such commodity, the issue of abrupt ban in agriculture commodities since 1939 until 2016 need urgent attention by the policy makers and regulators who wish to develop the commodity markets. The key issue in this regard

is does the price spikes in agri-commodities occur to due fundamentals like demand/supply factors or due destabilization effect of futures trading.

Based on the above mentioned issues , this paper attempts to delve deep into the dynamics of ban syndrome which has dripped Indian agriculture commodity derivative market since inception. The study examine all 14 commodities (tur, urad, sugar , wheat, rice , guarseed , guar gum, castor seed , rubber, chana , potato, soya refined oil) since 2003 to 2017 over which ban was imposed by exploring the impact of such abrupt suspension on spot return volatility and also study the price discovery mechanism spot and futures during pre-ban and post relaunch phases of these commodities. The study also attempts to take a relook at destabilization impact of futures trading on agriculture prices by examining the impact of unexpected trading activity of futures market (volume traded , open interest and value traded) using Hodrick Prescott filter

## **DATA & METHODOLOGY**

The sample data for daily volume , spot price , daily settlement price , open interest , value traded were obtained from NCDEX which generates 98% of total agriculture futures trading volume in India (SEBI, 2016). The list of 14 commodities (Table 1) that were suspended and relaunched were compiled from SEBI Circulars from its website. Since data for agriculture trading has been inflicted with bans and relaunched , Table 2 provides complete overview of period for which each commodity data has been available and corresponding spot market for which prices were obtained. Tur , Urad , Guarseed , Guar gum , Wheat , Soya refined Oil, Castor seed , Chana , Potato and Sugar were used for further analysis and since there were no or marginal trading volumes that too available for only 2 months for rubber and rice contracts, these two were excluded from sample. The futures price and corresponding average daily traded volumes were obtained from near month contracts leaving aside last week of trading for expiring contract due to excessive trading and physical settlement activity.

### **1. Process of Price Discovery using Cointegration**

The vast body of literature deploys Cointegration test for analyzing price discovery mechanism of futures markets (Chowdhury, 1991; Lai and Lai, 1991; Beck, 1994; Kellard et al., 1999; Yang et al., 2001; McKenzie and Holt, 2002; Kellard, 2002; Liu, 2004; Wang and Ke, 2005). The cointegration test enables to find whether a long run equilibrium exist between spot price and futures price which implies predictive signals of price convergence. Absence of cointegration reveals that futures have low information content of spot price and are not efficient. After exploring the long run relationship , nature and direction of short run relationship is also gauged using Granger Causality(p (Malliaris and Urrutia, 1998; Silvapulle and Moosa, 1999; Bryant et al., 2006).

As a pre-condition to cointegration test , the study first check stationarity of series using the standard methodology using conventional methods of Augmented Dickey Fuller (ADF) unit root test (Dickey and Fuller, 1979) and also cross checking with Phillips and Perron (PP) test. The spot prices and futures prices were converted to logarithmic returns  $\log(P_t/P_{t-1})$  and volume , open interest and value traded series if found non stationary were also taken in first differenced form.

Using Johansen and Juselius (1990) test , cointegration is used to explore the long run predictive power in spot price and futures price relationship . A n dimensional vector  $X_t$  with integration of order I(1) estimated vector autoregressive model by incorporating error correction as represented by following equation –

$$X_t = c + \sum_{i=1}^k \Pi_i X_{t-i} + \varepsilon_t$$

$$\Delta X_t = \mu + \sum_{i=1}^{k-1} \tau_i \Delta X_{t-i} + \Pi_k X_{t-k} + \varepsilon_t$$

Where  $X_t$  is a  $n \times 1$  vector of I(1) variable representing spot ( $S_t$ ) and futures ( $F_t$ ) and  $\mu$  represents deterministic component representing linear trend with  $\Pi_i$  represents  $n \times r$  matrix of parameters and  $k$  is lag length based on Schwarz criterion. The residual values of above model are used to construct trace test with null hypothesis of  $r$  cointegrating relation against alternative of  $k$  cointegrating relations.

After this, Granger causality test is applied which tells how much current value of one variable can be explained by past value of other variables and whether adding a lagged value of prior variables may improve explanatory power.

$$S_t = \alpha_0 + \alpha_i \sum_{i=0}^l F_{t-i} + \beta_j \sum_{j=0}^m S_{t-i} + u_t$$

$$F_t = \alpha'_0 + \alpha'_l \sum_{i=0}^l S_{t-i} + \beta'_j \sum_{j=0}^m F_{t-i} + v_t$$

Where  $S_t$  represent spot price and  $F_t$  represents futures price and granger causality enables us to gauge the short term relationship between the two and direction of causality as represented by the values of  $\alpha_i$  and  $\alpha'_l$  respectively.

## 2. Destabilisation Effect of Ban Syndrome using –

## 2.1. Augmented GARCH Model

Using the augmented GARCH methodology followed by (Working, 1962; Bessembinder and Seguin, 1992; Kamara, 1993; Antoniou *et al.*, 1998, 2005; Gulen and Mayhew, 2000; McKenzie *et al.*, 2001; Yang *et al.*, 2005) to test the impact of futures trading on spot return volatility, we deploy the same to explore the impact of ban and relaunch of commodity from futures trading by inputting BAN\_DUM as ban dummy and RELAUNCH\_DUM as relaunch dummy for given dates for 12 commodities both in mean and variance equation.

The spot market volatility has been calculated using EGARCH on spot returns of sample commodities using mean equation,

$$r_t = u_t + \beta_1 r_{t-1} + \log h_{t-1} + \hat{\beta}_1 BAN\_DUM + \hat{\beta}_2 RELAUNCH\_DUM + \varepsilon_t$$

The variance equation :

$$\log(h_t) = \gamma + \sum_{j=1}^q \varphi_j \left( \left| \frac{u_{t-j}}{\sqrt{h_{t-j}}} \right| \right) + \sum_{j=1}^q \phi_j \left( \frac{u_{t-j}}{\sqrt{h_{t-j}}} \right) + \sum_{i=1}^p \delta_i \log h_{t-i} + \beta_1 BAN\_DUM + \beta_2 RELAUNCH\_DUM$$

Where  $r_t$  is the return series of each sample commodity and  $\beta_1$  in the mean equation, and  $\gamma, \varphi_j, \phi_j, \delta_i$  in the variance equation are parameters to be estimated. The left-side variable is the log of the variance series. As it is well-known, the EGARCH model is usually applied to decipher the asymmetries in a series (Enders, 2010).

## 2.2. Unexpected Trading Activity on Spot Price Volatility

Another method to test the impact of ban and futures trading on spot return volatility is to examine the impact of unexpected trading activity of futures trading on spot return volatility. As proposed by (Bessembinder and Seguin, 1992; Darrat and Rahman, 1995; Yang *et al.*, 2005) unexpected liquidity measures were computed using high band pass filter with most widely adopted Hodrick and Prescott filter (HP, 1997). The rationale behind decomposing trading activity into expected and unexpected component is due to fact that expected or permanent component will be reflected in spot prices but unexpected trading represent shocks and abnormal trends not discernable in spot prices.

HP filter helps in decomposing a time series  $\{y_t\}$  into stochastic or permanent component,  $\mu_t$  and a stationary or transitory component,  $y_t - \mu_t$ . Real and nominal shocks bring about changes in a variable which can be segregated into trend and stationary components, respectively. Real shocks



cause permanent changes whereas nominal shocks have temporary effect on real prices. Permanent component  $y^*$  of a variable  $y$  is obtained by minimizing the following function:

$$\text{Min}\{y^*\} = \sum_{t=1}^T \frac{1(y_t - y^*)^2}{\sigma_0^2} + \frac{1}{\sigma_l^2} (\Delta^2 y^*)^2$$

$\sigma_0^2$  is the variance of stationary component  $y_t - y^*$ ,  $\sigma_l^2$  is the variance of growth rate of permanent or trend component (Basdevant, 2003). The technique allows for permanent or stochastic trend while deriving the stationary or temporary component. The cyclical or temporary component is derived by applying HP filter such that:

$$C = y - y^*$$

The unexpected liquidity components are computed for volume traded, open interest and value traded, then Granger causality has been used to test the direction of causality amongst unexpected liquidity component and spot return volatility. Using VAR framework, general form of granger causality can be specified as –

$$\begin{aligned} SV_t &= \alpha_0 + \alpha_i \sum_{i=0}^l SV_{t-i} + \beta_j \sum_{j=0}^m UL_{t-i} + u_t \\ UL_t &= \alpha'_0 + \alpha'_i \sum_{i=0}^l SV_{t-i} + \beta'_j \sum_{j=0}^m UL_{t-i} + v_t \end{aligned}$$

Where  $l$  and  $m$  are lag orders, and  $SV$  and  $UL$  are stationary time-series.  $SV$  denotes the spot market volatility and  $UL$  shows the unexpected liquidity (unexpected volumes, unexpected open interest and unexpected value traded). The regression residuals,  $u_t$  and  $v_t$ , are assumed to be i.i.d. The standard  $F$ -test or chi-square test is used to establish the direction of the causal relationship. The null hypothesis of this test is that  $UL$  does not Granger-cause  $SV$ . It is rejected when the coefficients of the lag values  $UL$  in equation (2) are jointly different from zero, i.e.  $\beta_1 \neq \beta_2 \neq \dots \neq \beta_n$

## DESCRIPTIVE ANALYSIS

Table 3 shows the descriptive statistics for spot return, future return, Volume, Open Interest and Value traded. The spot prices are positively skewed for most commodities but futures prices are negatively skewed. All return and trading activity variables are leptokurtic and are not normally distributed rejecting Jarque Bera test. Table 4 shows stationarity of above series and

spot prices and futures prices are stationary at first difference and volume and value are stationary at level.

## **EMPIRICAL ANALYSIS**

Table 5 shows Cointegration results to establish long run equilibrium amongst spot and futures prices and Table 6 depicts VAR adequacy measure like Modulus AR roots , Normality Test , Serial Correlation test to get stable results for VECM . Table 7 shows Vector error correction results and shows that  $\alpha$  is negative and significant for all commodities showing that long run equilibrium does exist for it . The wald test also came out to be rejected for all commodities except potato implying that short run equilibrium also exist for these commodities.. Table 8 shows the Granger causality test for spot return and futures return which further proves that futures lead the spot in all 12 commodities.

Table 9 shows the result of augmented GARCH model with ban and relaunch dummy variable and it was found that for wheat , chana , guarseed , guar gum , soya refined oil and sugar spot return volatility during ban phase was high and post relaunch phase fell which implies that futures market helps in reining in excess of market by rendering efficient price discovery and risk management. The post relaunch phase did lead more stable spot price volatility and price discovery also improved with speed of adjustment becoming high. This result was further corroborated by Table 10 that shows the mean and standard deviation of spot return volatility for three phases – pre-ban , during ban, post ban phase and it is found that wheat, sugar , guar gum , guarseed show excess average spot volatility during ban phase implying the beneficial role played by futures market using one way anova.

Table 11 gives extensive description of Granger Causality results of destabilization hypothesis and shows that post ban wheat , sugar , soya refined oil and guar gum depicted unexpected volumes and open interest causing spot price volatility and pre ban does not show such destabilization impact hinting to misguided ban decision of government as unexpected component did not impact spot volatility rather it was supply/demand imbalances.

## **CONCLUSION**

In this study, a novel attempt had been made to assess the impact of ban on price discovery , spot return volatility and destabilization hypothesis for Indian agriculture markets. The government is all up for the development of agriculture commodity markets but this study will help in making them realize that banning the commodity at drop of hat has done more harm than good by wiping away the trading volumes and trust from commixes. Speculators have not shown to rig the pre ban markets and spot volatility have been highest during ban phase implying the positive role played by futures market in reining the market excesses and infusing liquidity . The government should have better position limits and margin tracking mechanism to control speculative behavior rather than take the most extreme step of banning the commodity all together.

This study will show some light to policy makers that futures trading in agriculture will make price discovery mechanism more efficient and help in proper risk mitigation strategies.

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## APPENDIX

**TABLE 1- List of Commodities Suspended from Futures Trading as per SEBI Circulars (2003-2017)**

| S.No. | Commodities                       | Trading Suspended | Suspension Revoked |
|-------|-----------------------------------|-------------------|--------------------|
| 1     | Urad                              | 23 Jan, 2007      | Till Date          |
| 2     | Tur                               | 23 Jan, 2007      | 09-Feb-15          |
| 3     | Rice                              | 27 Feb, 2007      | Till Date          |
| 4     | Wheat                             | 27 Feb, 2007      | 21 May, 2009       |
| 5     | Chana , Soya Oil, Potato , Rubber | 08-May-08         | 04-Dec-08          |
| 6     | Sugar                             | 26 May, 2009      | 27-Dec-10          |
| 7     | Gaur Seed , Gaur Gum              | 27-Mar-12         | 14-May-13          |
| 8     | Potato                            | 30-Jul-12         | 01-Apr-13          |
| 9     | Castor Seeds                      | 27 Jan, 2016      | 05-Jan-17          |
| 10    | Chana                             | 27 July, 2016     | Till Date          |

**Source – SEBI Circulars**

**Table 2 – Details of Commodities selected with Spot market and Period Covered**

| COMMODITY        | SPOT MARKET | FUTURES EXCHANGE | PERIOD COVERED                           |
|------------------|-------------|------------------|------------------------------------------|
| TUR              | AKOLA       | NCDEX            | 4 MAY, 2006 TO 2 JANUARY , 2017          |
| URAD             | JALGAON     | NCDEX            | 18 JULY, 2006 TO 20 OCTOBER , 2016       |
| WHEAT            | DELHI       | NCDEX            | 18 JULY, 2005 TO 12 JUNE, 2013           |
| SUGAR            | KOLHPUR     | NCDEX            | 17 NOVEMBER, 2007 TO 11 SEPTEMBER , 2014 |
| SOYA REFINED OIL | INDORE      | NCDEX            | 17 DECEMBER , 2003 TO 10 DECEMBER , 2015 |
| CHANA            | DELHI       | NCDEX            | 13 APRIL, 2004 TO 27 JULY , 2016         |
| GUARSEED         | JODHPUR     | NCDEX            | 13 APRIL, 2004 TO 10 MAY, 2013           |
| GUARGUM          | JODHPUR     | NCDEX            | 27 JULY , 2004 TO 10 MAY 2013            |
| CASTORSEED       | DESSA       | NCDEX            | 20 JULY , 2009 TO 20 OCTOBER 2016        |
| POTATO           | DELHI       | NCDEX            | 5 MARCH , 2009 TO 12 SEPTEMBER , 2013    |

**TABLE 3 –**

| DESCRIPTIVE STATISTICS |                |             |                    |             |          |             |
|------------------------|----------------|-------------|--------------------|-------------|----------|-------------|
| COMMODITY              | VARIABLES      | MEAN        | STANDARD DEVIATION | SKEWNESS    | KURTOSIS | JARQUE-BERA |
| TUR                    | SPOT RETURN    | 0.00084472  | 0.006495876        | 1.111176747 | 6.002754 | 0           |
|                        | FUTURES RETURN | 0.00099066  | 0.008138435        | 0.326069304 | 4.471608 | 0.00058209  |
|                        | VOLUME         | 3860.072464 | 4215.814075        | 2.381129476 | 9.403973 | 0           |
|                        | OPEN INTEREST  | 21617.3913  | 14908.73292        | 1.285717747 | 4.133352 | 1.38E-10    |
|                        | VALUE TRADED   | 624399930.4 | 961836964.8        | 2.849537201 | 11.31336 | 0.00E+00    |
| URAD                   | SPOT RETURN    | 2.37E-05    | 0.008941821        | -0.64107874 | 8.846224 | 0           |
|                        | FUTURES RETURN | 0.000855    | 0.010541           | -0.160408   | 3.681742 | 0.161806    |
|                        | VOLUME         | 20968.9     | 17952              | 1.693266    | 7.507492 | 0           |
|                        | OPEN INTEREST  | 31024.48    | 16256.96           | -0.229443   | 2.354285 | 0.133547    |
|                        | VALUE TRADED   | 3.33E+09    | 3.14E+09           | 1.585574    | 5.902847 | 0           |
| Wheat(Pre-Ban)         | SPOT RETURN    | 0.000269455 | 0.005318645        | -2.91328462 | 29.29109 | 0           |
|                        | FUTURES RETURN | 0.000121468 | 0.005636224        | -1.99196233 | 25.26805 | 0           |
|                        | VOLUME         | 24672.35772 | 30254.94726        | 2.932545945 | 13.2079  | 0           |
|                        | OPEN INTEREST  | 246404.4106 | 121718.8705        | 1.464014536 | 4.731416 | 0           |
|                        | VALUE TRADED   | 1353524614  | 1404129596         | 2.46692916  | 10.37161 | 0           |
| Wheat(Post-Ban)        | SPOT RETURN    | 0.000136    | 0.003215           | 0.135109    | 13.96432 | 0           |
|                        | FUTURES RETURN | 0.000127    | 0.004596           | -1.000033   | 16.97394 | 0           |
|                        | VOLUME         | 4171.997    | 4942.431           | 5.189236    | 55.09622 | 0           |
|                        | OPEN INTEREST  | 37421.89    | 14856              | -0.033766   | 1.794298 | 0           |
|                        | VALUE TRADED   | 2.44E+08    | 2.68E+08           | 5.922139    | 74.51314 | 0           |
| SUGAR(Pre-Ban)         | SPOT RETURN    | 0.000535582 | 0.004289162        | 2.31794542  | 20.84591 | 0           |
|                        | FUTURES RETURN | 0.000624661 | 0.005663963        | 1.241444297 | 8.053282 | 0           |
|                        | VOLUME         | 16214.69799 | 14337.2269         | 2.849540367 | 17.91264 | 0           |
|                        | OPEN           | 143186.3982 | 32875.74933        | 0.524220576 | 2.84215  | 2.84E-05    |



|                                   |                   |            |            |             |          |          |
|-----------------------------------|-------------------|------------|------------|-------------|----------|----------|
|                                   | INTEREST          |            |            |             |          |          |
|                                   | VALUE<br>TRADED   | 1873114784 | 1582507007 | 2.036637147 | 9.13099  | 0        |
| SUGAR(Post-Ban)                   | SPOT<br>RETURN    | 1.18E-05   | 0.002509   | 1.848549    | 15.48026 | 0        |
|                                   | FUTURES<br>RETURN | -1.99E-05  | 0.003644   | 0.403426    | 7.517609 | 0        |
|                                   | VOLUME            | 8553.058   | 7994.911   | 1.987779    | 8.600692 | 0        |
|                                   | OPEN<br>INTEREST  | 63152.64   | 21729.9    | 0.076312    | 2.977356 | 0.589367 |
|                                   | VALUE<br>TRADED   | 1.14E+09   | 9.77E+08   | 2.275735    | 10.98373 | 0        |
| SOYA<br>REFINED<br>OIL(Pre-Ban)   | SPOT<br>RETURN    | 7.75E-05   | 0.003006   | -0.359678   | 13.45262 | 0        |
|                                   | FUTURES<br>RETURN | 7.56E-05   | 0.003812   | -0.062705   | 6.242782 | 0        |
|                                   | VOLUME            | 14998.18   | 14746.72   | 1.821692    | 7.395127 | 0        |
|                                   | OPEN<br>INTEREST  | 60751.57   | 37377.39   | 0.109011    | 2.090866 | 0        |
|                                   | VALUE<br>TRADED   | 3.12E+09   | 3.11E+09   | 2.462392    | 14.10207 | 0        |
| SOYA<br>REFINED OIL<br>(Post-Ban) | SPOT<br>RETURN    | 6.51E-05   | 0.003408   | 0.2281      | 10.22802 | 0        |
|                                   | FUTURES<br>RETURN | 1.62E-05   | 0.005233   | -5.742664   | 129.5885 | 0        |
|                                   | VOLUME            | 67413.45   | 52838.28   | 1.451044    | 5.206643 | 0        |
|                                   | OPEN<br>INTEREST  | 221761.8   | 65344.27   | -0.436184   | 3.042947 | 0        |
|                                   | VALUE<br>TRADED   | 1.90E+10   | 1.37E+10   | 1.319743    | 4.676011 | 0        |
| CHANA (Pre-Ban)                   | SPOT<br>RETURN    | 0.000171   | 0.006129   | -0.397562   | 8.288479 | 0        |
|                                   | FUTURES<br>RETURN | 0.000184   | 0.006717   | -1.105088   | 13.36392 | 0        |
|                                   | VOLUME            | 65504.59   | 85362.03   | 2.989326    | 15.10807 | 0        |
|                                   | OPEN<br>INTEREST  | 169650     | 98601.73   | 0.804907    | 3.498672 | 0        |
|                                   | VALUE<br>TRADED   | 9.72E+09   | 9.34E+09   | 1.384088    | 5.011368 | 0        |
| CHANA (Post-Ban)                  | SPOT<br>RETURN    | 0.000259   | 0.006022   | 0.657266    | 6.839985 | 0        |
|                                   | FUTURES<br>RETURN | 0.000236   | 0.007021   | -0.657973   | 11.40296 | 0        |
|                                   | VOLUME            | 73779.3    | 55184.49   | 1.48256     | 6.770897 | 0        |
|                                   | OPEN              | 206630     | 109110.4   | 0.749151    | 4.350296 | 0        |

|                         |                   |             |             |             |          |   |
|-------------------------|-------------------|-------------|-------------|-------------|----------|---|
|                         | INTEREST          |             |             |             |          |   |
|                         | VALUE<br>TRADED   | 9.97E+09    | 7.41E+09    | 1.860718    | 8.461565 | 0 |
| GUARSEED<br>(Pre-Ban)   | SPOT<br>RETURN    | 0.000579    | 0.00774     | -0.299712   | 16.94963 | 0 |
|                         | FUTURES<br>RETURN | 0.000563    | 0.008645    | -0.549845   | 16.68823 | 0 |
|                         | VOLUME            | 101091.5    | 99252.15    | 1.404829    | 4.786573 | 0 |
|                         | OPEN<br>INTEREST  | 224869.6    | 71633.44    | -1.058881   | 4.867422 | 0 |
|                         | VALUE<br>TRADED   | 1.47E+10    | 1.12E+10    | 1.676044    | 7.474552 | 0 |
|                         |                   |             |             |             |          |   |
| GUARGUM<br>(Pre- Ban)   | SPOT<br>RETURN    | 0.000555534 | 0.0082376   | -0.45283233 | 14.57622 | 0 |
|                         | FUTURES<br>RETURN | 0.000520892 | 0.008786803 | -0.8094284  | 14.5908  | 0 |
|                         | VOLUME            | 5266.429752 | 6167.430318 | 1.98243149  | 8.381333 | 0 |
|                         | OPEN<br>INTEREST  | 33064.82253 | 17542.39147 | 0.391432075 | 1.854309 | 0 |
|                         | VALUE<br>TRADED   | 2091492470  | 2680745277  | 2.717693751 | 15.50835 | 0 |
|                         |                   |             |             |             |          |   |
| CASTORSEED<br>(Pre-Ban) | SPOT<br>RETURN    | 7.41E-05    | 0.006529    | -0.228521   | 12.73453 | 0 |
|                         | FUTURES<br>RETURN | 5.43E-05    | 0.00749     | -0.098958   | 5.72002  | 0 |
|                         | VOLUME            | 33431.49    | 41850.08    | 1.349295    | 4.352949 | 0 |
|                         | OPEN<br>INTEREST  | 155002.2    | 147088.9    | 0.457919    | 1.879897 | 0 |
|                         | VALUE<br>TRADED   | 5.84E+09    | 6.81E+09    | 1.371929    | 4.774986 | 0 |
|                         |                   |             |             |             |          |   |
| POTATO (Pre-<br>Ban)    | SPOT<br>RETURN    | 0.000418919 | 0.020336263 | 1.774063735 | 308.1976 | 0 |
|                         | FUTURES<br>RETURN | 0.000439802 | 0.024192805 | 3.609285901 | 183.7656 | 0 |
|                         | VOLUME            | 12557       | 16020.58188 | 2.398672632 | 11.16072 | 0 |
|                         | OPEN<br>INTEREST  | 52436.20833 | 53816.29477 | 1.217017487 | 3.586525 | 0 |
|                         | VALUE<br>TRADED   | 185661224.1 | 186303152.8 | 1.790657488 | 6.613787 | 0 |
|                         |                   |             |             |             |          |   |

**TABLE 4 –**

| STATIONARITY TEST |           |       |      |       |      |       |      |       |
|-------------------|-----------|-------|------|-------|------|-------|------|-------|
|                   |           | ADF   |      |       |      | PP    |      |       |
| COMMODITY         | VARIABLES | LEVEL | PVAL | FIRST | PVAL | LEVEL | PVAL | FIRST |

|                     |               | T STAT    |        | DIFFERENCE |        |          |        | DIFFERENCE |
|---------------------|---------------|-----------|--------|------------|--------|----------|--------|------------|
| TUR                 | SPOT PRICE    | -1.91768  | 0.3243 | -47.61974  | 0.0001 | -1.90928 | 0.3282 | -47.64657  |
|                     | FUTURES PRICE | -1.129384 | 0.7031 | -11.97966  | 0      | -1.24272 | 0.6547 | -11.97717  |
|                     | VOLUME        | -5.513699 | 0      | -9.590722  | 0      | -5.3846  | 0      | -28.79704  |
|                     | OPEN INTEREST | 0.627011  | 0.99   | -5.229983  | 0      | 1.483157 | 0.9992 | -10.17109  |
|                     | VALUE TRADED  | 0.944548  | 0.9959 | -13.41932  | 0      | -1.03168 | 0.7408 | -13.62915  |
| URAD                | SPOT PRICE    | -1.089482 | 0.7221 | -53.1512   | 0.0001 | -1.02239 | 0.7473 | -53.1504   |
|                     | FUTURES PRICE | -1.748992 | 0.4047 | -9.709471  | 0      | -1.63876 | 0.4604 | -9.691512  |
|                     | VOLUME        | -3.84236  | 0.0031 | -11.33469  | 0      | -5.25445 | 0      | -26.59756  |
|                     | OPEN INTEREST | -1.851569 | 0.3545 | -9.847946  | 0      | -1.85639 | 0.3522 | -10.18863  |
|                     | VALUE TRADED  | -2.028804 | 0.2743 | -6.381371  | 0      | -3.89524 | 0.0026 | -27.81577  |
| WHEAT<br>(Pre ban)  | SPOT PRICE    | -1.504865 | 0.5305 | -17.32584  | 0      | -1.41897 | 0.5737 | -17.29704  |
|                     | FUTURES PRICE | -1.332216 | 0.6158 | -20.87157  | 0      | -1.37179 | 0.5968 | -20.85074  |
|                     | VOLUME        | -7.597202 | 0      | -13.6624   | 0      | -9.79716 | 0      | -54.38575  |
|                     | OPEN INTEREST | -1.456068 | 0.5552 | -10.75495  | 0      | -1.41344 | 0.5764 | -18.02827  |
|                     | VALUE TRADED  | -5.727428 | 0      | -11.04705  | 0      | -8.73311 | 0      | -56.34083  |
| WHEAT<br>(Post Ban) | SPOT PRICE    | -1.149533 | 0.6978 | -27.63336  | 0      | -1.07801 | 0.7263 | -28.1372   |
|                     | FUTURES PRICE | -1.116165 | 0.7114 | -35.08536  | 0      | -1.20647 | 0.6737 | -35.12337  |
|                     | VOLUME        | -8.38084  | 0      | -23.90037  | 0      | -16.9352 | 0      | -85.9567   |
|                     | OPEN INTEREST | -2.261368 | 0.185  | -15.71703  | 0      | -2.44606 | 0.1294 | -30.44462  |
|                     | VALUE TRADED  | -7.84581  | 0      | -18.44883  | 0      | -18.7211 | 0      | -105.9902  |
| SUGAR<br>(Pre ban)  | SPOT PRICE    | 0.277771  | 0.9253 | -17.46486  | 0      | -0.25458 | 0.9285 | -17.78316  |
|                     | FUTURES PRICE | 0.333092  | 0.9171 | -20.90933  | 0      | -0.29487 | 0.9228 | -20.90838  |
|                     | VOLUME        | 4.476837  | 0.0002 | -26.74817  | 0      | -12.3485 | 0      | -101.3399  |
|                     | OPEN          | -         | 0.0877 | -17.62871  | 0      | -2.90429 | 0.0456 | -18.25383  |

|                                   |               |            |        |           |        |          |        |           |
|-----------------------------------|---------------|------------|--------|-----------|--------|----------|--------|-----------|
|                                   | INTEREST      | 2.629581   |        |           |        |          |        |           |
|                                   | VALUE TRADED  | - 4.982964 | 0      | -22.32529 | 0      | -9.00311 | 0      | -63.85505 |
| SUGAR<br>(Post Ban)               | SPOT PRICE    | - 1.466517 | 0.5504 | -27.1401  | 0      | -1.63028 | 0.4667 | -28.03351 |
|                                   | FUTURES PRICE | - 2.014298 | 0.2808 | -17.63132 | 0      | -2.00262 | 0.2859 | -30.24265 |
|                                   | VOLUME        | -6.75712   | 0      | -22.83313 | 0      | -19.4471 | 0      | -84.65784 |
|                                   | OPEN INTEREST | - 2.882185 | 0.0478 | -23.57385 | 0      | -3.19152 | 0.0208 | -24.13356 |
|                                   | VALUE TRADED  | - 5.381126 | 0      | -22.90065 | 0      | -19.362  | 0      | -85.80073 |
|                                   |               |            |        |           |        |          |        |           |
| SOYA<br>REFINED OIL<br>(Pre ban)  | SPOT PRICE    | - 0.294355 | 0.9233 | -27.1445  | 0      | -0.48561 | 0.8914 | -27.49141 |
|                                   | FUTURES PRICE | - 0.399131 | 0.9069 | -32.98971 | 0      | -0.38763 | 0.9088 | -33.01451 |
|                                   | VOLUME        | - 5.336356 | 0      | -17.81927 | 0      | -15.5454 | 0      | -115.6047 |
|                                   | OPEN INTEREST | - 1.846745 | 0.3579 | -33.70163 | 0      | -1.94028 | 0.3138 | -33.75557 |
|                                   | VALUE TRADED  | - 3.693804 | 0.0043 | -25.15656 | 0      | -14.675  | 0      | -98.40833 |
|                                   |               |            |        |           |        |          |        |           |
| SOYA<br>REFINED OIL<br>(Post Ban) | SPOT PRICE    | - 1.771593 | 0.395  | -33.22228 | 0      | -1.7885  | 0.3866 | -33.6929  |
|                                   | FUTURES PRICE | - 1.831838 | 0.3652 | -50.48682 | 0.0001 | -1.91482 | 0.3256 | -50.09738 |
|                                   | VOLUME        | - 3.401745 | 0.011  | -19.48559 | 0      | -23.9758 | 0      | -126.8316 |
|                                   | OPEN INTEREST | -2.97658   | 0.0373 | -41.71075 | 0      | -2.90833 | 0.0446 | -41.64105 |
|                                   | VALUE TRADED  | - 3.794885 | 0.003  | -26.74998 | 0      | -21.0493 | 0      | -125.9122 |
|                                   |               |            |        |           |        |          |        |           |
| CHANA<br>(Pre ban)                | SPOT PRICE    | -1.58673   | 0.4891 | -31.6968  | 0      | -1.48611 | 0.5406 | -31.53514 |
|                                   | FUTURES PRICE | - 1.460793 | 0.5534 | -34.33525 | 0      | -1.47706 | 0.5452 | -34.34118 |
|                                   | VOLUME        | - 7.495909 | 0      | -19.85375 | 0      | -14.5858 | 0      | -64.70053 |
|                                   | OPEN INTEREST | - 2.108189 | 0.2416 | -30.31992 | 0      | -2.1663  | 0.219  | -30.55812 |
|                                   | VALUE TRADED  | - 2.934613 | 0.0418 | -14.214   | 0      | -9.31677 | 0      | -104.6439 |
|                                   |               |            |        |           |        |          |        |           |
| CHANA<br>(Post Ban)               | SPOT PRICE    | 2.482258   | 1      | -33.01    | 0      | 2.333043 | 1      | -40.73524 |
|                                   | FUTURES PRICE | 1.6907     | 0.9997 | -44.98899 | 0.0001 | 1.767805 | 0.9997 | -44.98919 |
|                                   | VOLUME        | -3.15359   | 0.023  | -18.78687 | 0      | -22.3486 | 0      | -120.9226 |

|                                       |               |           |        |           |        |          |        |           |
|---------------------------------------|---------------|-----------|--------|-----------|--------|----------|--------|-----------|
|                                       | OPEN INTEREST | -2.054099 | 0.2638 | -39.06989 | 0      | -2.37576 | 0.1488 | -40.18908 |
|                                       | VALUE TRADED  | -5.412566 | 0      | -31.23639 | 0      | -24.6835 | 0      | -168.1502 |
| GUARSEED                              | SPOT PRICE    | 7.537516  | 1      | -42.71912 | 0      | 19.81398 | 1      | -43.53978 |
|                                       | FUTURES PRICE | 7.011369  | 1      | -44.8412  | 0.0001 | 13.3119  | 1      | -45.60389 |
|                                       | VOLUME        | -10.88804 | 0      | -19.67857 | 0      | -20.5462 | 0      | -152.7507 |
|                                       | OPEN INTEREST | -2.845204 | 0.0522 | -48.30145 | 0.0001 | -2.92818 | 0.0423 | -48.31358 |
|                                       | VALUE TRADED  | -5.503502 | 0      | -35.16011 | 0      | -24.2299 | 0      | -192.9212 |
| GUARGUM                               | SPOT PRICE    | 7.964945  | 1      | -42.05113 | 0      | 23.9782  | 1      | -43.3222  |
|                                       | FUTURES PRICE | 7.468268  | 1      | -43.71541 | 0      | 14.89947 | 1      | -44.91405 |
|                                       | VOLUME        | -9.136941 | 0      | -19.75661 | 0      | -20.7992 | 0      | -206.4318 |
|                                       | OPEN INTEREST | -1.348888 | 0.6086 | -42.28422 | 0      | -1.48092 | 0.5434 | -42.80782 |
|                                       | VALUE TRADED  | -3.703359 | 0.0041 | -22.38349 | 0      | -22.8721 | 0      | -226.4743 |
| CASTORSEED                            | SPOT PRICE    | -2.552395 | 0.1034 | -28.55903 | 0      | -2.66389 | 0.0806 | -32.36759 |
|                                       | FUTURES PRICE | -2.785853 | 0.0605 | -33.99353 | 0      | -2.75409 | 0.0653 | -33.83524 |
|                                       | VOLUME        | -2.402447 | 0.1412 | -17.17607 | 0      | -12.7463 | 0      | -117.3382 |
|                                       | OPEN INTEREST | -0.29157  | 0.9237 | -21.43109 | 0      | -0.47035 | 0.8943 | -34.38239 |
|                                       | VALUE TRADED  | -2.69803  | 0.0745 | -14.4718  | 0      | -11.3855 | 0      | -188.7195 |
| POTATO                                | SPOT PRICE    | -1.268902 | 0.6456 | -25.366   | 0      | -1.34524 | 0.6099 | -25.36655 |
|                                       | FUTURES PRICE | -1.036165 | 0.7418 | -25.86227 | 0      | -1.10818 | 0.7143 | -25.86317 |
|                                       | VOLUME        | -2.546016 | 0.1051 | -12.31689 | 0      | -6.34518 | 0      | -38.22187 |
|                                       | OPEN INTEREST | -2.331485 | 0.1623 | -7.013079 | 0      | -2.02637 | 0.2755 | -22.51906 |
|                                       | VALUE TRADED  | -4.545482 | 0.0002 | -19.56519 | 0      | -10.8601 | 0      | -58.30717 |
| *MacKinnon (1996) one-sided p-values. |               |           |        |           |        |          |        |           |

**TABLE 5 –**

| COINTEGRATION EQUATIONS |                        |                  |         |
|-------------------------|------------------------|------------------|---------|
| COMMODITY               | Hypotheses             | Trace Statistics | P-Value |
| TUR                     | No Cointegration       | 12.90651         | 0.1183  |
|                         | Atmost 1 Cointegration | 0.132305         | 0.716   |
| URAD                    | No Cointegration       | 16.42249         | 0.0362  |
|                         | Atmost 1 Cointegration | 3.04421          | 0.081   |
| WHEAT(PRE-BAN)          | No Cointegration       | 10.21149         | 0.2648  |
|                         | Atmost 1 Cointegration | 1.800258         | 0.1797  |
| WHEAT(POST-BAN)         | No Cointegration       | 50.32848         | 0       |
|                         | Atmost 1 Cointegration | 1.402996         | 0.2362  |
| SUGAR (PRE-BAN)         | No Cointegration       | 22.17921         | 0.0042  |
|                         | Atmost 1 Cointegration | 0.017724         | 0.894   |
| SUGAR (POST-BAN)        | No Cointegration       | 45.63722         | 0       |
|                         | Atmost 1 Cointegration | 2.076387         | 0.1496  |
| SOYA REF OIL (PRE-BAN)  | No Cointegration       | 37.87369         | 0       |
|                         | Atmost 1 Cointegration | 0.291817         | 0.5891  |
| SOYA REF OIL (POST-BAN) | No Cointegration       | 68.17024         | 0       |
|                         | Atmost 1 Cointegration | 3.624375         | 0.0569  |
| CHANA (PRE-BAN)         | No Cointegration       | 28.75891         | 0.0003  |
|                         | Atmost 1 Cointegration | 2.216724         | 0.1365  |
| CHANA (POST-BAN)        | No Cointegration       | 71.22373         | 0       |
|                         | Atmost 1 Cointegration | 5.18704          | 0.0227  |
| GUARSEED                | No Cointegration       | 301.6932         | 0.0001  |
|                         | Atmost 1 Cointegration | 96.45865         | 0       |
|                         | Atmost 2 Cointegration |                  |         |
| GUARGUM                 | No Cointegration       |                  |         |
|                         | Atmost 1 Cointegration |                  |         |
| CASTORSEED              | No Cointegration       | 58.08547         | 0       |
|                         | Atmost 1 Cointegration | 6.640189         | 0.01    |
|                         | Atmost 2 Cointegration |                  |         |
| POTATO                  | No Cointegration       | 20.53254         | 0.008   |
|                         | Atmost 1 Cointegration | 2.657992         | 0.103   |

**TABLE 6 –**

| VAR ADEQUACY TEST |                                     |                 |         |
|-------------------|-------------------------------------|-----------------|---------|
| COMMODITY         | TESTS                               | Test Statistics | P-Value |
| URAD              | VAR Residual Normality Tests(Jarque | 153.6983        | 0       |

|                       |                                                                         |                                              |        |
|-----------------------|-------------------------------------------------------------------------|----------------------------------------------|--------|
|                       | bera)                                                                   |                                              |        |
|                       | VAR Residual Serial Correlation LM Tests                                | 4.314205                                     | 0.3651 |
|                       | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.972697<br>0.875134<br>0.304594<br>0.018477 | NA     |
| WHEAT(POST-BAN)       | VAR Residual Normality Tests(Jarque bera)                               | 30253.1                                      | 0      |
|                       | VAR Residual Serial Correlation LM Tests                                | 7.63564                                      | 0.1059 |
|                       | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.996520<br>0.952450<br>0.190592<br>0.080229 |        |
| SUGAR (PRE-BAN)       | VAR Residual Normality Tests(Jarque bera)                               | 4624.228                                     | 0      |
|                       | VAR Residual Serial Correlation LM Tests                                | 4.98782                                      | 0.2885 |
|                       | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.999605<br>0.876409<br>0.166959<br>0.166959 |        |
| SUGAR (POST-BAN)      | VAR Residual Normality Tests(Jarque bera)                               | 5031.833                                     | 0      |
|                       | VAR Residual Serial Correlation LM Tests                                | 4.525368                                     | 0.3396 |
|                       | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.996322<br>0.907984<br>0.196301<br>0.196301 |        |
| SOYA REF OIL(PRE-BAN) | VAR Residual Normality Tests(Jarque bera)                               | 22102.49                                     | 0      |
|                       | VAR Residual Serial Correlation LM Tests                                | 16.38643                                     | 0.0025 |
|                       | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.998758<br>0.947308<br>0.248123<br>0.244550 |        |

|                               |                                                                            |                                                                      |        |
|-------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------|--------|
| SOYA REF<br>OIL(POST-<br>BAN) | VAR Residual Normality Tests(Jarque<br>bera)                               | 7193.564                                                             | 0      |
|                               | VAR Residual Serial Correlation LM<br>Tests                                | 10.96345                                                             | 0.027  |
|                               |                                                                            | 0.997250<br>0.939141<br>0.222533<br>0.157998                         |        |
|                               | VAR STABILITY Test -Modulus value<br>of Roots of Characteristic Polynomial |                                                                      |        |
| CHANA (PRE-<br>BAN)           | VAR Residual Normality Tests(Jarque<br>bera)                               | 12278.68                                                             | 0      |
|                               | VAR Residual Serial Correlation LM<br>Tests                                | 4.692561                                                             | 0.3203 |
|                               |                                                                            | 0.996783<br>0.964971<br>0.200528<br>0.061079                         |        |
|                               | VAR STABILITY Test -Modulus value<br>of Roots of Characteristic Polynomial |                                                                      |        |
| CHANA(POST-<br>BAN)           | VAR Residual Normality Tests(Jarque<br>bera)                               | 51205.87                                                             |        |
|                               | VAR Residual Serial Correlation LM<br>Tests                                | 1.827132                                                             | 0.7675 |
|                               |                                                                            | 1.002788<br>0.934418<br>0.319200<br>0.319200<br>0.260758<br>0.260758 |        |
|                               | VAR STABILITY Test -Modulus value<br>of Roots of Characteristic Polynomial |                                                                      |        |
| GUARSEED                      | VAR Residual Normality Tests(Jarque<br>bera)                               | 16096636                                                             |        |
|                               | VAR Residual Serial Correlation LM<br>Tests                                | 7.608681                                                             | 0.107  |



|            |                                                                         |                                                                                                                                                                      |        |
|------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|            |                                                                         | 1.026339<br>1.026339<br>1.018030<br>0.996154<br>0.996154<br>0.965260<br>0.965260<br>0.894229<br>0.812766<br>0.812766<br>0.636957<br>0.617876<br>0.617876<br>0.092985 |        |
|            | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial |                                                                                                                                                                      |        |
| CASTORSEED | VAR Residual Normality Tests(Jarque bera)                               | 14306.07                                                                                                                                                             |        |
|            | VAR Residual Serial Correlation LM Tests                                | 8.558591                                                                                                                                                             | 0.0731 |
|            | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.992477<br>0.929005<br>0.382546<br>0.382546<br>0.156769<br>0.156769                                                                                                 |        |
| POTATO     | VAR Residual Normality Tests(Jarque bera)                               | 4572727                                                                                                                                                              | 0      |
|            | VAR Residual Serial Correlation LM Tests                                | 4.183926                                                                                                                                                             | 0.3817 |
|            | VAR STABILITY Test -Modulus value of Roots of Characteristic Polynomial | 0.986302<br>0.977869<br>0.029077<br>0.029077                                                                                                                         |        |

**TABLE 7**

| VECM RESULTS                |     |                       |                           |                           |       |                           |                           |       |                   |
|-----------------------------|-----|-----------------------|---------------------------|---------------------------|-------|---------------------------|---------------------------|-------|-------------------|
| COMMODITY                   |     | A                     | ΔSt-1                     | ΔSt-2                     | ΔSt-3 | ΔFt-1                     | ΔFt-2                     | ΔFt-3 | WALD TEST (CHISQ) |
| URAD                        | ΔFt | -0.027150<br>[0.4886] | 0.053327<br>[0.6294]      | 0.181174<br>[0.0897]      |       | 0.204413<br>[0.0464]      | -<br>0.106157<br>[0.3297] |       |                   |
|                             | ΔSt | -0.113689<br>[0.0012] | 0.073383<br>[0.4521]      | 0.124483<br>[0.1847]      |       | 0.373905<br>[0.0001]      | -<br>0.111837<br>[0.2416] |       | 0                 |
| WHEAT (POST BAN)            | ΔFt | -0.002369<br>[0.8268] | -<br>0.052693<br>[0.1050] | -<br>0.049962<br>[0.1227] |       | 0.113738<br>[0.0124]      | 0.017310<br>[0.6956]      |       |                   |
|                             | ΔSt | 0.046867<br>[0.000]   | 0.068993<br>[0.0019]      | -<br>0.016906<br>[0.4441] |       | 0.170321<br>[0.0000]      | -<br>0.053587<br>[0.0762] |       | 0                 |
| SUGAR (PRE-BAN)             | ΔFt | -0.031968<br>[0.2765] | 0.034278<br>[0.5488]      | -<br>0.062394<br>[0.2810] |       | -<br>0.065838<br>[0.3732] | 0.049411<br>[0.4581]      |       |                   |
|                             | ΔSt | 0.074531<br>[0.0005]  | 0.240165<br>[0.0000]      | 0.045309<br>[0.2777]      |       | 0.066381<br>[0.2133]      | -<br>0.054385<br>[0.2576] |       | 0                 |
| SUGAR (POST-BAN)            | ΔFt | -0.059566<br>[0.0002] | 0.133940<br>[0.0001]      | -<br>0.041356<br>[0.2510] |       | -<br>0.087510<br>[0.0826] | 0.056489<br>[0.2121]      |       |                   |
|                             | ΔSt | 0.037194<br>[0.0003]  | 0.279810<br>[0.0000]      | 0.030609<br>[0.1922]      |       | 0.036424<br>[0.2672]      | -<br>0.003591<br>[0.9031] |       | 0                 |
| SOYA REFINED OIL (PRE-BAN)  | ΔFt | 0.000823<br>[0.9639]  | 0.009888<br>[0.8125]      | -<br>0.013152<br>[0.7581] |       | 0.196707<br>[0.0003]      | -<br>0.026286<br>[0.5778] |       |                   |
|                             | ΔSt | 0.056939<br>[0.0000]  | 0.344742<br>[0.0000]      | 0.062465<br>[0.0455]      |       | -<br>0.046517<br>[0.2416] | -<br>0.032664<br>[0.3439] |       | 0                 |
| SOYA REFINED OIL (POST-BAN) | ΔFt | -0.011173<br>[0.4225] | -<br>0.040885<br>[0.1903] | -<br>0.034156<br>[0.2738] |       | 0.133383<br>[0.0031]      | 0.051842<br>[0.1791]      |       |                   |

|                     |             |                       |                        |                        |                          |                        |                        |                       |            |
|---------------------|-------------|-----------------------|------------------------|------------------------|--------------------------|------------------------|------------------------|-----------------------|------------|
|                     | $\Delta St$ | 0.050676<br>[0.0000]  | 0.242262<br>[0.0000]   | 0.017551<br>[0.3921]   |                          | 0.076558<br>[0.0096]   | 0.025653<br>[0.3116]   |                       | 0          |
| CHANA<br>(PRE-BAN)  | $\Delta Ft$ | -0.00659<br>[0.5862]  | 0.000346<br>[0.9921]   | -0.02547<br>[0.5050]   |                          | 0.034622<br>[0.4023]   | 0.033839<br>[0.3387]   |                       |            |
|                     |             |                       |                        | -                      |                          | -                      |                        |                       |            |
|                     | $\Delta St$ | 0.040141<br>[0.0000]  | 0.436074<br>[0.0000]   | 0.097231<br>[0.0018]   |                          | 0.085234<br>[0.0114]   | 0.060374<br>[0.0361]   |                       | 0          |
| CHANA(P<br>OST-BAN) | $\Delta Ft$ | -0.051249<br>[0.0006] | 0.07864<br>[0.0076]    | - 0.060529<br>[0.0623] | 0.01087<br>5<br>[0.7247] | 0.101099<br>[0.0023]   | - 0.000705<br>[0.9832] | 0.022192<br>[0.4477]  |            |
|                     |             |                       |                        |                        | 0.01487<br>1<br>[0.5745] | - 0.178538<br>[0.0000] | - 0.110587<br>[0.0001] | -0.006115<br>[0.8073] |            |
|                     | $\Delta St$ | 0.046363<br>[0.0003]  | 0.477443<br>[0.0000]   | 0.01022<br>[0.7135]    |                          |                        |                        |                       |            |
| CASTORSE<br>ED      | $\Delta Ft$ | 0.016180<br>[0.3519]  | - 0.010787<br>[0.7970] | - 0.078926<br>[0.0378] |                          | 0.155641<br>[0.0000]   | - 0.000626<br>[0.9867] |                       |            |
|                     |             |                       | -                      | -                      |                          |                        |                        |                       |            |
|                     | $\Delta St$ | -0.071545<br>[0.0000] | 0.079987<br>[0.0206]   | 0.172787<br>[0.0000]   |                          | 0.372934<br>[0.0000]   | 0.042308<br>[0.1701]   |                       |            |
| POTATO              | $\Delta Ft$ | -0.043285<br>[0.0240] | 0.011277<br>[0.8883]   |                        |                          | 0.010049<br>[0.8926]   |                        |                       |            |
|                     |             |                       |                        |                        |                          | -                      |                        |                       |            |
|                     | $\Delta St$ | -0.066224<br>[0.0002] | 0.045666<br>[0.5349]   |                        |                          | 0.022011<br>[0.7469]   |                        |                       | 0.754<br>0 |

**TABLE 8 –**

| SPOT-FUTURES SHORT RUN CAUSALITY |                                                   |        |        |
|----------------------------------|---------------------------------------------------|--------|--------|
| COMMODITY                        | GRANGER CAUSALITY                                 | F stat | P val  |
| TUR                              | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 12.047 | 0.0007 |
|                                  | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 0.4923 | 0.4841 |

|                         |                                                   |         |           |
|-------------------------|---------------------------------------------------|---------|-----------|
| URAD                    | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 8.68828 | 0.0003    |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 0.98393 | 0.3763    |
| WHEAT(Pre-Ban)          | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 19.1617 | 1.00E-08  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 1.40004 | 0.2476    |
| WHEAT(Post-Ban)         | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 23.7879 | 1.00E-06  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 6.1255  | 0.0135    |
| SUGAR (Pre-Ban)         | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 55.7586 | 4.00E-13  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 0.85696 | 0.3551    |
| SUGAR (Post-Ban)        | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 183.607 | 1.00E-38  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 4.67356 | 0.0309    |
| SOYA REF OIL(Pre-Ban)   | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 152.023 | 4.00E-33  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 10.8793 | 0.001     |
| SOYA REF OIL (Post-Ban) | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 134.722 | 4.00E-30  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 7.46246 | 0.0064    |
| CHANA (Pre-Ban)         | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 312.713 | 2.00E-62  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 0.18925 | 0.6636    |
| CHANA (Post-Ban)        | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 622.37  | 9.00E-121 |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 1.08384 | 0.298     |
| GUARSEED                | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 111.498 | 2.00E-67  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 13.4138 | 1.00E-08  |
| GUARGUM                 | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 109.681 | 1.00E-85  |
|                         | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 7.2366  | 9.00E-06  |
| CASTORSEED              | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 1.46226 | 0.232     |

|        |                                                   |         |          |
|--------|---------------------------------------------------|---------|----------|
|        | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 116.879 | 6.00E-48 |
|        | FUTURES_RETURN does not Granger Cause SPOT_RETURN | 0.54547 | 0.5798   |
| POTATO | SPOT_RETURN does not Granger Cause FUTURES_RETURN | 0.02153 | 0.9787   |

**TABLE 9 –**

| COMMODITY           | VARIABLES      | Coefficients | p-value     |
|---------------------|----------------|--------------|-------------|
| TUR                 | BAN_DUMMY      | 7.85E-08     | 0.845811171 |
|                     | RELAUNCH_DUMMY | 6.77E-06     | 4.95E-10    |
| URAD                | BAN_DUMMY      | 0            | 1           |
| WHEAT               | BAN_DUMMY      | 6.88E-07     | 0.0188      |
|                     | RELAUNCH_DUMMY | -7.96E-07    | 0           |
| SUGAR               | BAN_DUMMY      | 1.26E-06     | 0           |
|                     | RELAUNCH_DUMMY | -8.04E-07    | 0           |
| SOYA<br>REFINED OIL | BAN_DUMMY      | 3.77E-07     | 0.0006      |
|                     | RELAUNCH_DUMMY | 1.99E-08     | 0.451       |
| CHANA               | BAN_DUMMY      | 3.37E-08     | 0.7942      |
|                     | RELAUNCH_DUMMY | 1.86E-07     | 0.0049      |
| GUARSEED            | BAN_DUMMY      | 6.26E-06     | 0.002       |
| GUARGUM             | BAN_DUMMY      | 3.87E-06     | 0.0003      |
| CASTORSEED          | BAN_DUMMY      | -2.62E-06    | 0           |
| POTATO              | BAN_DUMMY      | 0            | 1           |
|                     | RELAUNCH_DUMMY | 0            | 1           |

**TABLE 10 –**

| EQUALITY OF MEANS TEST(t test/Anova) |                                      |      |              |                |                |         |  |
|--------------------------------------|--------------------------------------|------|--------------|----------------|----------------|---------|--|
| COMMODITY                            | PHASES                               | N    | Mean         | Std. Deviation | T/F statistics | P-value |  |
| TUR                                  | PRE- BAN SPOT<br>RETURN VOLATILITY   | 138  | .00004368578 | .000033476009  | -6.065         | .000    |  |
|                                      | DURING BAN SPOT<br>RETURN VOLATILITY | 2582 | .00006315130 | .000075046573  |                |         |  |
| URAD                                 | PRE- BAN SPOT<br>RETURN VOLATILITY   | 154  | .0000237     | .00894182      | -.166          | .868    |  |
|                                      | DURING BAN SPOT<br>RETURN VOLATILITY | 2596 | .0001359     | .00810010      |                |         |  |
| WHEAT                                | PRE- BAN SPOT                        | 492  | .000036741   | .0000890702    | 46.029         | .000    |  |

|                     |                                      |      |              |               |         |      |
|---------------------|--------------------------------------|------|--------------|---------------|---------|------|
|                     | RETURN VOLATILITY                    |      |              |               |         |      |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 659  | .000029088   | .0000265261   |         |      |
|                     | POST-BAN SPOT<br>RETURN VOLATILITY   | 1204 | .000014305   | .0000283028   |         |      |
| SUGAR               | PRE- BAN SPOT<br>RETURN VOLATILITY   | 447  | .0000204842  | .00002431531  | 297.660 | .000 |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 469  | .0000348919  | .00003050200  |         |      |
|                     | POST-BAN SPOT<br>RETURN VOLATILITY   | 1067 | .0000078619  | .00001036654  |         |      |
| SOYA<br>REFINED OIL | PRE- BAN SPOT<br>RETURN VOLATILITY   | 1297 | .00000909227 | .000007497424 | 115.573 | .000 |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 167  | .00002086802 | .000020574693 |         |      |
|                     | POST-BAN SPOT<br>RETURN VOLATILITY   | 1964 | .00001099209 | .000009160477 |         |      |
| CHANA               | PRE- BAN SPOT<br>RETURN VOLATILITY   | 1233 | .0000358905  | .00002658326  | 12.877  | .000 |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 168  | .0000268685  | .00000961900  |         |      |
|                     | POST-BAN SPOT<br>RETURN VOLATILITY   | 2139 | .0000364551  | .00002249537  |         |      |
| GUARSEED            | PRE- BAN SPOT<br>RETURN VOLATILITY   | 2382 | .00006169348 | .000087321347 | -21.405 | 0    |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 321  | .00036959298 | .000255722413 |         |      |
| GUARGUM             | PRE- BAN SPOT<br>RETURN VOLATILITY   | 2299 | .00007004229 | .000076055872 | -23.051 | .000 |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 321  | .00048620476 | .000322216337 |         |      |
| CASTORSEED          | PRE- BAN SPOT<br>RETURN VOLATILITY   | 1540 | .00003870197 | .000032199231 | 7.144   | .000 |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 158  | .00002867785 | .000014307437 |         |      |
| POTATO              | PRE- BAN SPOT<br>RETURN VOLATILITY   | 720  | .00041316458 | .007245260437 | .324    | .723 |
|                     | DURING BAN SPOT<br>RETURN VOLATILITY | 88   | .00001968524 | .000037100081 |         |      |
|                     | POST-BAN SPOT<br>RETURN VOLATILITY   | 135  | .00002363432 | .000139480762 |         |      |

**TABLE 11 –**

| DESTABILISATION HYPOTHESIS RESULTS |                                                         |         |          |               |
|------------------------------------|---------------------------------------------------------|---------|----------|---------------|
| COMMODITY                          | PAIR OF VARIABLES                                       | F- Test | P-Val    | Granger Cause |
| TUR (pre-ban)                      | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 1.98248 | 0.161445 | No            |
|                                    | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 7.42255 | 0.0073   | Yes           |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 1.31374 | 0.2538   | No            |
|                                    | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 3.76507 | 0.0544   | No            |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 5.73904 | 0.001    | Yes           |
|                                    | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 14.612  | 3E-08    | Yes           |
| URAD (pre-ban)                     | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 6.66004 | 0.0108   | No            |
|                                    | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 0.85899 | 0.3555   | No            |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 2.01599 | 0.1577   | No            |
|                                    | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.33647 | 0.5627   | No            |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 0.13067 | 0.7182   | No            |
|                                    | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 0.00449 | 0.9467   | No            |
| WHEAT (Pre-Ban)                    | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 2.84475 | 0.0591   | No            |
|                                    | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 21.4803 | 1E-09    | Yes           |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 2.24443 | 0.1071   | No            |
|                                    | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.59495 | 0.552    | No            |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 0.32532 | 0.7225   | No            |
|                                    | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 2.38418 | 0.0932   | No            |
| WHEAT (Post-Ban)                   | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 26.4522 | 5E-21    | Yes           |
|                                    | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 21.8702 | 2E-17    | Yes           |
|                                    | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 5.72063 | 0.0034   | Yes           |

|                                   |                                                         |         |          |     |
|-----------------------------------|---------------------------------------------------------|---------|----------|-----|
|                                   | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 2.76897 | 0.0631   | No  |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 40.7268 | 5E-32    | Yes |
|                                   | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 28.8151 | 7E-23    | Yes |
| SUGAR<br>(Pre-Ban)                | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 8.66134 | 0.00001  | Yes |
|                                   | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 7.38769 | 0.00008  | Yes |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 0.45634 | 0.4997   | No  |
|                                   | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.3036  | 0.5819   | No  |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 2.47188 | 0.0612   | No  |
|                                   | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 35.2245 | 2E-20    | Yes |
|                                   |                                                         |         |          |     |
| SUGAR<br>(Post-Ban)               | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 11.5593 | 3E-09    | Yes |
|                                   | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 3.81358 | 0.0044   | Yes |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 3.44215 | 0.016321 | Yes |
|                                   | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 3.17916 | 0.0233   | Yes |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 13.7407 | 9E-09    | Yes |
|                                   | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 35.3301 | 9E-22    | Yes |
|                                   |                                                         |         |          |     |
| Soya refined oil<br>(Pre-Ban)     | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 0.58625 | 0.6726   | No  |
|                                   | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 7.0351  | 0.00001  | Yes |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 1.05092 | 0.3861   | No  |
|                                   | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 2.34791 | 0.0391   | Yes |
|                                   | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 2.24065 | 0.0372   | Yes |
|                                   | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 8.30441 | 8.00E-09 | Yes |
|                                   |                                                         |         |          |     |
| SOYA<br>REFINED OIL<br>(Post-Ban) | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 1.12002 | 0.3474   | No  |
|                                   | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 1.62313 | 0.1243   | No  |



|                     |                                                         |         |          |     |
|---------------------|---------------------------------------------------------|---------|----------|-----|
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 1.65609 | 0.1912   | No  |
|                     | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.15636 | 0.8553   | No  |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 0.7993  | 0.5878   | No  |
|                     | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 5.20418 | 7.00E-06 | Yes |
| CHANA<br>(Pre-Ban)  | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 0.60718 | 0.6575   | No  |
|                     | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 2.28508 | 0.0583   | No  |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 1.7651  | 0.152    | No  |
|                     | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 1.29993 | 0.273    | No  |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 0.69357 | 0.4051   | No  |
|                     | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 0.24417 | 0.6213   | No  |
| CHANA<br>(Post-Ban) | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 1.568   | 0.2087   | No  |
|                     | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 8.60486 | 0.0002   | Yes |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 2.21619 | 0.1367   | No  |
|                     | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.48001 | 0.4885   | No  |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 0.7756  | 0.541    | No  |
|                     | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 10.4173 | 2.00E-08 | Yes |
| GUARSEED            | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 0.14597 | 0.9323   | No  |
|                     | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 1.20234 | 0.3074   | No  |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 11.3379 | 1.00E-05 | Yes |
|                     | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.5191  | 0.5951   | No  |
|                     | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 3.35321 | 0.0027   | Yes |
|                     | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 2.79282 | 0.0104   | Yes |
| GUARGUM             | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 0.18101 | 0.8344   | No  |

|            |                                                         |          |        |     |
|------------|---------------------------------------------------------|----------|--------|-----|
|            | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 7.18578  | 0.0008 | Yes |
|            | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 1.23206  | 0.2671 | No  |
|            | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.26831  | 0.6045 | No  |
|            | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 3.22482  | 0.0037 | Yes |
|            | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 4.59499  | 0.0001 | Yes |
| CASTORSEED | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 1.39903  | 0.2414 | No  |
|            | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 4.00895  | 0.0074 | Yes |
|            | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 0.14636  | 0.7021 | No  |
|            | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 7.2441   | 0.0072 | Yes |
|            | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 4.25558  | 0.0053 | Yes |
|            | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 4.71673  | 0.0028 | Yes |
| POTATO     | SPOT_VOLATILITY does not causes UNEXPECTED_VOLUMES      | 0.5621   | 0.5703 | No  |
|            | UNEXPECTED_VOLUMES does not causes SPOT_VOLATILITY      | 2.20E-05 | 1      | No  |
|            | SPOT_VOLATILITY does not causes UNEXPECTED_OPENINTEREST | 3.25477  | 0.0392 | No  |
|            | UNEXPECTED_OPENINTEREST does not causes SPOT_VOLATILITY | 0.02908  | 0.9713 | No  |
|            | SPOT_VOLATILITY does not causes UNEXPECTED_VALUE        | 2.87307  | 0.0572 | No  |
|            | UNEXPECTED_VALUE does not causes SPOT_VOLATILITY        | 0.72068  | 0.4868 | No  |